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4.2.5. Titanic Dataset Analysis and Data Cleaning

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You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using .info()).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the ‘Age’ column with the median age.
8. Fill missing values in the ‘Embarked’ column with the most frequent value (mode()).
9. Drop the ‘Cabin’ column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embark

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd

import numpy as np


# Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset

print(data.head())


# 2. Display the last 5 rows of the dataset

print(data.tail())


# 3. Get the shape of the dataset

print(data.shape)


# 4. Get a summary of the dataset (info)

print(data.info())


# 5. Get basic statistics of the dataset

print(data.describe())


# 6. Check for missing values

print(data.isnull().sum())
```

7. Fill missing values in the ‘Age’ column with the median age

data['Age'] = data['Age'].fillna(data['Age'].median())

8. Fill missing values in the ‘Embarked’ column with the mode

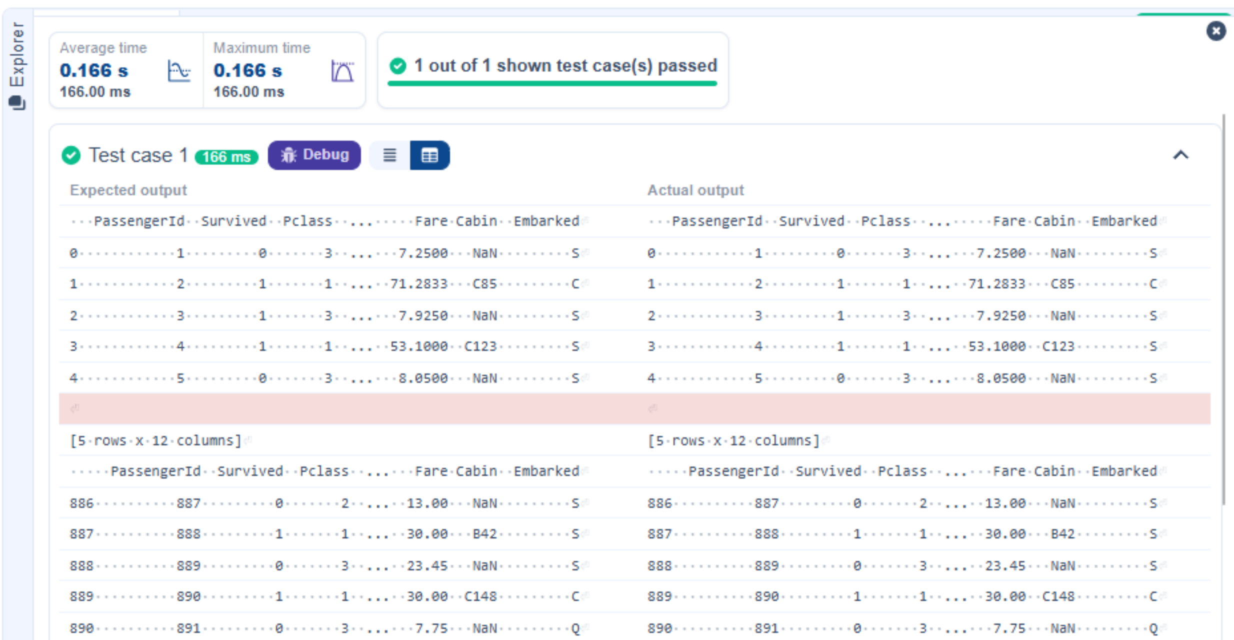
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])

9. Drop the ‘Cabin’ column due to many missing values

data = data.drop(columns=['Cabin'])

10. Create a new column 'FamilySize’ by adding ‘SibSp' and ‘Parch'

data['FamilySize'] = data['SibSp'] + data['Parch']



Explorer

Average time **0.166 s** 166.00 ms

Maximum time **0.166 s** 166.00 ms

1 out of 1 shown test case(s) passed

Debugger

[5 rows x 12 columns]	[5 rows x 12 columns]
(891, 12)	(891, 12)
<class 'pandas.core.frame.DataFrame'>	<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890	RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):	Data columns (total 12 columns):
# Column Non-Null Count Dtype	# Column Non-Null Count Dtype
--- ------ -------------- -----	--- ------ -------------- -----
0 PassengerId 891 non-null int64	0 PassengerId 891 non-null int64
1 Survived 891 non-null int64	1 Survived 891 non-null int64
2 Pclass 891 non-null int64	2 Pclass 891 non-null int64
3 Name 891 non-null object	3 Name 891 non-null object
4 Sex 891 non-null object	4 Sex 891 non-null object
5 Age 714 non-null float64	5 Age 714 non-null float64
6 SibSp 891 non-null int64	6 SibSp 891 non-null int64
7 Parch 891 non-null int64	7 Parch 891 non-null int64
8 Ticket 891 non-null object	8 Ticket 891 non-null object
9 Fare 891 non-null float64	9 Fare 891 non-null float64
10 Cabin 204 non-null object	10 Cabin 204 non-null object
11 Embarked 889 non-null object	11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)	dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB	memory usage: 66.2+ KB
None	None
PassengerId Survived Pclass ... SibSp Parch Fare	PassengerId Survived Pclass ... SibSp Parch Fare
count 891.000000 891.000000 891.000000 ... 891.000000 891.000000 891.000000	count 891.000000 891.000000 891.000000 ... 891.000000 891.000000 891.000000
mean 446.000000 0.383838 2.308642 ... 0.523008 0.381594 32.204208	mean 446.000000 0.383838 2.308642 ... 0.523008 0.381594 32.204208
std 257.353842 0.486592 0.836071 ... 1.102743 0.806057 49.693429	std 257.353842 0.486592 0.836071 ... 1.102743 0.806057 49.693429

Terminal Test cases

Explorer

Average time
0.166 s
166.00 ms

Maximum time
0.166 s
166.00 ms

1 out of 1 shown test case(s) passed

```
count   891.000000  891.000000  891.000000  ...   891.000000  891.00
0000  891.000000
mean    446.000000    0.383838    2.308642  ...     0.523008    0.38
1594   32.204208
std     257.353842    0.486592    0.836071  ...     1.102743    0.80
6057   49.693429
min       1.000000    0.000000    1.000000  ...     0.000000    0.00
0000    0.000000
25%     223.500000    0.000000    2.000000  ...     0.000000    0.00
0000    7.910400
50%     446.000000    0.000000    3.000000  ...     0.000000    0.00
0000   14.454200
75%     668.500000    1.000000    3.000000  ...     1.000000    0.00
0000   31.000000
max     891.000000    1.000000    3.000000  ...     8.000000    6.00
0000  512.329200

[8 rows x 7 columns]
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```


```
count   891.000000  891.000000  891.000000  ...   891.000000  891.00
0000  891.000000
mean    446.000000    0.383838    2.308642  ...     0.523008    0.38
1594   32.204208
std     257.353842    0.486592    0.836071  ...     1.102743    0.80
6057   49.693429
min       1.000000    0.000000    1.000000  ...     0.000000    0.00
0000    0.000000
25%     223.500000    0.000000    2.000000  ...     0.000000    0.00
0000    7.910400
50%     446.000000    0.000000    3.000000  ...     0.000000    0.00
0000   14.454200
75%     668.500000    1.000000    3.000000  ...     1.000000    0.00
0000   31.000000
max     891.000000    1.000000    3.000000  ...     8.000000    6.00
0000  512.329200

[8 rows x 7 columns]
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```


Debugger

Terminal | Test cases

‹ Prev | Reset | Submit | Next ›

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4/24/2026